NETWORKING

DHCP
Visual Guide

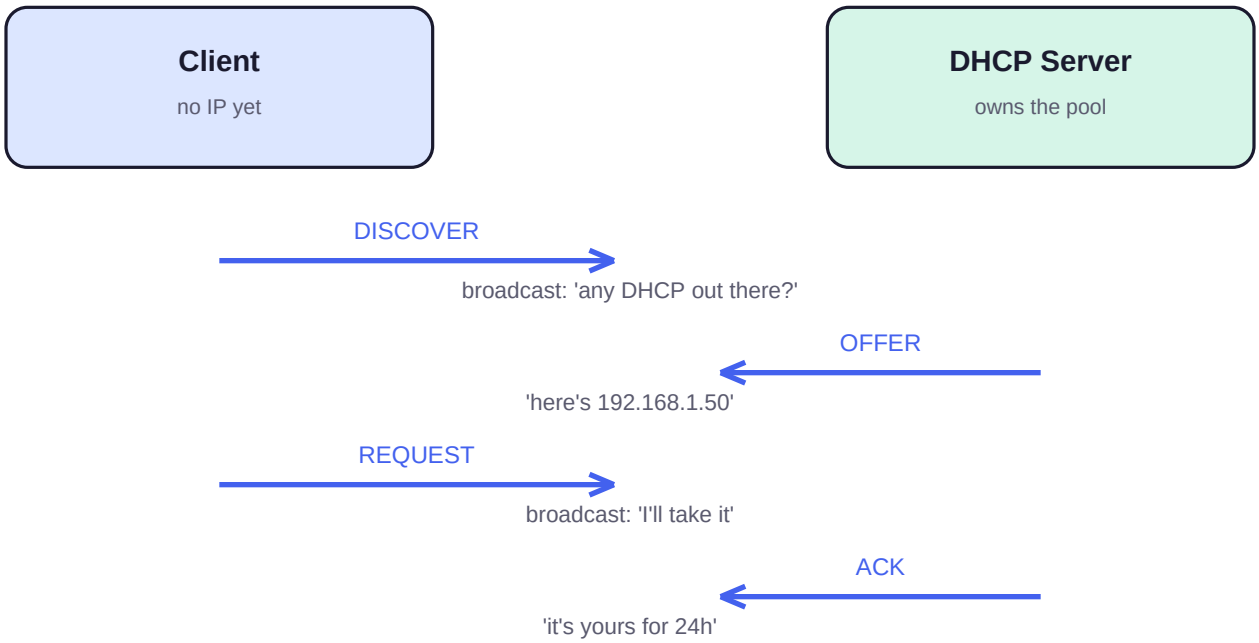
Dynamic Host Configuration —
how devices get an IP without anyone typing one.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

DORA: The Four-Step Dance

Discover → Offer → Request → Acknowledge.



Early messages are BROADCASTS (255.255.255.255) — the client has no IP to be reached at yet.

STEP 2/5

Leases: Nothing Is Forever

IPs are rented, not owned — with renewal timers built in.

Lease granted: 24 hours

T1 = 12h (50%) → try to RENEW with original server

T2 = 21h (87.5%)

no answer? REBIND — broadcast to ANY server

Expiry

lease lapses → stop using the IP, restart DORA

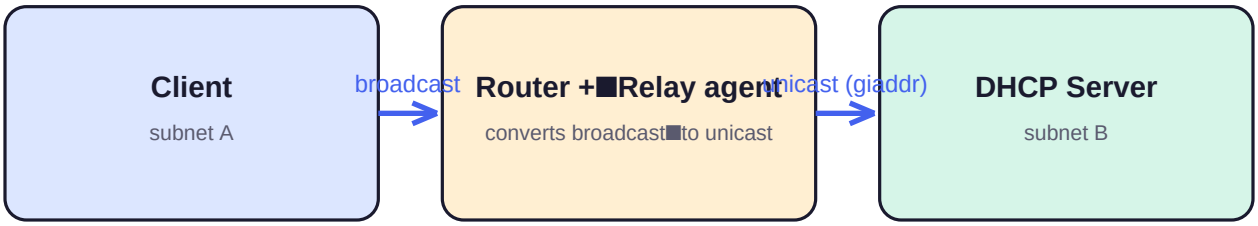
Renewal is graceful: the client keeps its address across renewals — users never notice.

Short leases (hours) for guest Wi-Fi and hot-desks; long leases (days/weeks) for stable desktops and printers.

STEP 3/5

Relay Agents: Crossing Subnets

Broadcasts don't cross routers — so the router forwards for you.



- The relay stamps the giaddr field with the client's subnet so the server picks the right pool.
- One central DHCP server can serve dozens of VLANs — no server per subnet needed.
- 'ip helper-address' on the router interface is the classic config line.

STEP 4/5

DHCP Options: More Than an IP

The ACK carries the whole network bootstrap kit.

Option 3 — Router

default gateway: the way out of the subnet

Option 6 — DNS

name servers the client should query

Option 51 — Lease time

how long the address is valid

Option 66/67 — Boot

TFTP server + boot file for PXE booting

Option 15 — Domain

DNS search suffix for short names

Options are TLV (type-length-value) — extensible without changing the protocol.

STEP 5/5

Rogue Servers & Hardening

DHCP trusts whoever answers first — that's the attack.

- Rogue DHCP: an attacker answers DISCOVER first and hands out THEIR gateway — classic man-in-the-middle setup.
- DHCP snooping: switches allow DHCP replies only on trusted uplink ports; client-port replies are dropped.
- Starvation: attacker requests every address in the pool so legit clients get nothing — snooping + rate limits help.
- Reservations: bind critical hosts (printers, servers) to fixed IPs by MAC — still managed centrally.

DHCPv6 exists but IPv6 leans on SLAAC for address assignment — DHCPv6 mainly delivers DNS and other options.

STEP ✓/✓

Key Takeaways

DHCP — Remember

- DORA: Discover (broadcast) → Offer → Request → Ack — the client starts with no IP.
- Leases expire; clients renew at 50% (T1) and rebind at 87.5% (T2).
- Relay agents forward broadcasts across subnets with giaddr stamped in.
- Options deliver gateway, DNS, lease time, boot info — a full bootstrap kit.
- Harden with DHCP snooping: trust only uplink ports for server replies.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev